Swagat Bhattacharyya

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SUMMARY

With a strong foundation in electrical engineering, applied physics, and mathematics, my research portfolio spans neuromorphic systems, mixed-signal design, and signal processing. My master's work at Georgia Tech focused on developing neuromorphic computing systems and culminated in several papers and tapeouts across several tech nodes. My industrial experiences, including roles at Renesas and Neurava, have provided skills in robust embedded systems development and entrepreneurship, translating ideas into impactful products. I am now researching photoacoustic tomography at Caltech.

EDUCATION

California Institute of Technology	Pasadena, CA | --
Pursuing a PhD in Medical and Electrical Engineering	**GPA: --**
Georgia Institute of Technology	Atlanta, GA | July 2024
Master of Science in Electrical Engineering | *Advisor: Prof. Jennifer O. Hasler*	**GPA: 3.90**
Purdue University, Honors College	West Lafayette, IN | May 2022
Triple Major: *BS Electrical Engineering, BS Applied Physics, & BS Mathematics*	**GPA: 3.96** (Highest Distinction)

PUBLICATIONS (H-Index: 6; Citations: 64 According to Google Scholar: https://tinyurl.com/yc66j52v)

Patents	1	Irazoqui P, Shah J, Meyer T, Ganesh V, **Bhattacharyya S**, &Hsiung Y, "Multimodal Seizure Sensing"
Journals (peer-reviewed)	1	***Bhattacharyya S*, "**Performance Bounds for a Maxima-Sampling Envelope Detector," IEEE Transactions on Circuits and Systems II: Express Briefs, vol 72, no 10, Oct 2025
	2	***Bhattacharyya S*** &Hasler J, "A Six-Transistor Integrate-and-Fire Neuron Enabling Chaotic Dynamics," IEEE Transactions on Biomedical Circuits and Systems, vol 19, no 5, Oct 2025
	3	Kurup S, ***Bhattacharyya S***, …, &Vogel EM, "Modular HfO_x-RRAM for On-Demand Micromodular Electronics," IEEE Electron Device Letters, vol 46, no 7, July 2025
	4	Mathews P, Ayyappan PR, Ige A, ***Bhattacharyya S***, Yang L, &Hasler J, "A 65 nm CMOS Analog Programmable Standard Cell Library for Mixed-Signal Computing," IEEE Transactions on VLSI Syst., vol 32, no 10, Oct 2024
	5	***Bhattacharyya S*** &Hasler J, "Extrema-Triggered Conversion for Non-Stationary Signal Acquisition in Wireless Sensor Nodes," JLPEA, vol 14, no 1, Feb 2024
	6	***Bhattacharyya S***, Ayyappan PR, &Hasler J, "Towards Scalable Digital Modeling of Networks of Biorealistic Silicon Neurons," IEEE Journal on Emerging and Selected Topics in Circuits and Systems, vol 13, no 4, Dec 2023
	7	***Bhattacharyya S*** &Graham DW, "Amplitude-Regulated Quadrature Sine-VCO Employing an OTA-C Topology," IEEE Transactions on Circuits and Systems II: Express Briefs, vol 70, no 6, June 2023
	8	***Bhattacharyya S***, Andryzcik S, &Graham DW, "An Acoustic Vehicle Detector & Classifier Using a Reconfigurable Analog/Mixed-Signal Platform," JLPEA, vol 10, no 1, Article 6, March-April, 2020
Confer-ence Talks	1	Mathews P, Ayyappan PR, Ige A, ***Bhattacharyya S***, Yang L, & Hasler J, "A 65nm and 130nm CMOS program-mable analog standard cell library for scalable system synthesis," IEEE CICC, Denver, CO, Apr 21-24, 2024
	2	***Bhattacharyya S***, Yang L, & Hasler J, "BuzzSort: A Linear-Time, Event-Driven Data Conversion and Sorting Framework for Approximate Computing Architectures," IEEE ICRC, San Diego, CA, Dec 5-6, 2023
	3	***Bhattacharyya S*** & Hasler J, "Extrema-Triggered Analog-Digital Conversion for Low-Power Wireless Sensor Nodes," IEEE MWSCAS, Tempe, AZ, Aug 6-9, 2023
	4	***Bhattacharyya S***, Mathews P, Ayyappan PR, & Hasler J, "Toward Biorealistic Silicon Neural Circuits on Reconfigurable Platforms," IEEE MWSCAS, Tempe, AZ, Aug 6-9, 2023
Posters (selected)	1	Meyer T, **Bhattacharya S**, Lehman P, Ganesh V, Ta J, Lowen K, Dragon D, Sainju R, Gehlbach B, Shah J, Nobis W, & Richerson G, "A Novel Multi-Modal Arm Wearable for Seizure Detection," AES Conf., Orlando, FL, Dec 1-5, 2023
	2	Rumple M, **Bhattacharyya S**, Hagedorn I, & Ghera Z, "RevEx: Low-Power, Nonoptical VR Limb Tracking with Eddy-Current Haptics," Purdue Spark Challenge, West Lafayette, IN, Dec 10, 2020
	3	Prakash M, Szadowski H, Thompson M, Dextre A, Chan M, Lee J, **Bhattacharyya S**, Ravichandran K, Saylor D, &Howard B, "Microfluidic Argonaute Mediated COVID-19 Diagnostic Device," Fall Research Expo, West Lafayette, IN, Nov 16-20, 2020
	4	**Bhattacharyya S**, "DIMOS: A Low-Power, Fast Response Logic Gate Architecture," Intel ISEF, Pittsburgh, May 13-18, 2018

HONORS & AWARDS (SELECTED)

Selected for the Caltech Division of Engineering & Applied Science (EAS) Chair Scholars Program	2026
National Science Foundation Graduate Research Fellowship (nationwide acceptance rate: ~15%)	2022
IEEE International Conference on Rebooting Computing (ICRC) Best Short Paper Award	2023
Georgia Institute of Technology Presidential Fellowship	2022

Purdue Mathematics Department Gordon L. Walker Scholarship	2021
Purdue Physics and Astronomy Department Scholarship	2021
2nd place in the Purdue School of ECE Spark Challenge (senior design showcase)	2021
2nd place in Embedded Systems and 1st place NSA Research Directorate Award in Computing at Intel ISEF – honored with a minor planet in my name: 34619 Swagat	2018
Purdue Trustees Scholarship (Waiver for Half of Tuition)	2018

ACADEMIC RESEARCH EXPERIENCE [KEY: ● => Hardware | ○ => Software]

Eng. & Applied Sci. Chair Scholar, Caltech Optical Imaging Laboratory, *Caltech* *Sept. 2026 –*

- ● Researched Photoacoustic Tomography.

NSF Graduate Research Fellow, Integrated Computational Electronics Lab, *Georgia Tech* *Aug. 2022 – July 2024*

- ● Designed, laid out, and **taped out *three chips*** using floating-gate (FG) FETs in *28 nm, 180 nm,* and *350 nm* CMOS; Ensured robust performance of the following **analog standard cells** across process nodes: 6-bit DAC, FG-based mixer, FG bootstrap source, I&F neurons, strongARM latch, and high-voltage decoders for FG programming.
- ● Successfully experimentally demonstrated signal conditioning circuits using glass-transferred RRAM devices.
- ● **Characterized cryogenic FG FETs** and developed robust FG FET programming methods in *65 nm* CMOS.
- ● Led a team demonstrating the first **analog sorting algorithm** (award-winning talk at IEEE ICRC 2023).
- ● Developed a **nonuniform ADC** leveraging extrema sampling for data reduction given a non-stationary input.
- ● Invented and demonstrated the simplest non-driven **chaotic CMOS oscillator** (a 6T FG-based I&F neuron).
- ○ Led a team developing the first efficient C, Matlab, &Python models of transistor-based **HH neuron networks**.
- ○ Led a team developing a fast simulation suite in Python for large-scale **analog architectural exploration**.

Undergraduate Researcher, Computational Electronic Systems Lab, *West Virginia Univ.* *June 2014 – June 2022*

- ● Developed a quadrature, **OTA-C sine VCO** with the widest reported tuning range via a novel **envelope detector**.
- ● Designed a low-power in-sole **gait analyzer** to study the progression of Parkinson's Disease in a nonclinical setting.
- ○ Developed **analog ML** algorithms to train an acoustic vehicle classifier on a field-programmable analog array.
- ○ Developed **mixed-integer nonlinear programming algorithms** to optimize starved logic gate and ADC topologies.

Undergraduate Researcher, Webb Group, Dept. of ECE, *Purdue University* Jan. 2022 – May 2022

- ● Fabricated 3D-printed phantoms to quantify the scatterer density estimation resolution in speckle imaging.
- ○ Developed fast cross-correlation algorithms for high-res speckle images that could run on both GPUs and CPUs.

Undergraduate Researcher, Center for Implantable Devices, *Purdue University* *Jan. 2021 – May 2021*

- ● Designed, simulated, and characterized a high-order filter for a novel implantable gastric monitoring sensor.

INDUSTRIAL EXPERIENCE [KEY: ●=> Hardware | ○=> Software | ❖ => Business Development]

Field Applications Engineer, Renesas Electronics America, *Raleigh, NC & Boston, MA* *Aug. 2024 – Sept. 2026*

- ○ Developed and delivered a 2-hr, hands-on **technical workshop** for our distribution FAE counterparts showing how to develop stand-alone **server applications** on Renesas's new Wi-Fi 6 product line (Montreal, Oct. 2025).
- ○ Developed custom software and provided technical support for Renesas's WiFi 6 and RA MCU lineups.
- ○ Authored an application note demonstrating **high-throughput DSP** implementation on the RA8 product line, including measurements of performance and power draw characterization across various test conditions.
- ❖ Drove new design-ins and platform adoption across industrial, medical, and consumer embedded applications, growing Renesas's New England funnel and revenue in embedded processing and wireless connectivity.

Cofounder & COO, Prody.Zone, Nashua, NH Mar. 2024 - Current

- ○ Codesigned a web-based coworking application to help people stay productive while working from home.
- ❖ Led customer discovery interviews, usability studies, field testing, and the launch.

Cofounder & Embedded Design Engineer, Lune Systems, *Atlanta, GA* *Oct. 2023 – Mar. 2026*

- ● Led **PCB design &testing** of a cost-effective smart insomnia aid, focused on optimal, user-centric performance.
- ● Conducted in-depth customer discovery to define **product specifications** as tailored to market needs &trends.
- ○ Engineered robust embedded firmware to enhance the performance and the user interface of the aid.

Hardware Design Engineer, Neurava LLC, *West Lafayette, IN* *May 2021 – July 2022*

- ● Led a team of five in creating a **seizure sensor array (SSA)** for SUDEP-risk patients from concept through initial clinical trials overseeing: biomarker&sensor selection; data pipeline; PCB design, layout, fabrication, &testing.
- ❖ Conducted the initial patent search and cowrote the **patent** application for the multi-modal SSA. The SSA, which has undergone clinical testing since 2022, achieved the **lowest-known false positive rate** in the industry while maintaining state-of-the-art sensitivity, directly contributing towards securing **$3M** in seed funding for Neurava LLC.
- ○ Developed embedded SSA firmware (DSP &controls) for real-time, low-power **mixed-signal edge computing**.
- ○ Proposed, developed, &implemented **automated characterization protocols** to ensure SSA *reliability*.

OTHER LEADERSHIP EXPERIENCES

Publicity Director, NH IEEE Section — 2026
- Advertised NH IEEE and its events on platforms like LinkedIn / Eventbrite & in-person.

President, Purdue IEEE Student Branch — 2020-21
- Oversaw operations and bylaw compliance of 11 committees with ~200 members in total.
- Spearheaded outreach events & initiated several intra- and inter-student branch collabs.

Vice President, Purdue IEEE Student Branch — 2019-20
- Oversaw operations of six technical committees, managed shared resources, improved workspace safety and compliance with protocol, & resolved inter-committee conflicts.

Electrical Design & Fabrication Lead, Purdue IEEE Engineering in Medicine & Biology Society — 2019-20, 21-22
- Taught 20 team members skills to design, prototype, and fabricate circuits during my tenure.
- Designed and constructed a navigational aid for the visually impaired.
- Designed and constructed the feedback control systems for an exoskeletal assistive arm.

SERVICES (Selected)

I. Peer Review

Journal Name	Number of Papers Reviewed
IEEE Transactions on Biomedical Circuits and Systems (TBioCAS)	4
IEEE Journal on Emerging and Selected Topics in Circuits and Systems (JETCAS)	1
IEEE Midwest Symposium on Circuits and Systems (MWSCAS)	3

II. Mentoring / Tutoring

StudentU | Durham, NC — Sep-Dec, 2024
- Tutored highschoolers from marginalized minority communities in Durham, NC for 2 hours per week in math and chemistry, fostering a supportive environment for learning.

ICE Lab | Atlanta, Georgia — Aug-Dec, 2023
- Closely mentored an undergraduate neuroscience student at Georgia Tech in a semester-long research project involving simulations of biorealistic CMOS neuron networks.

III. Community Outreach (Selected)

NH IEEE Outreach Seminar — Feb. 12, 2026
- Gave a seminar at Nashua Community College to increase interest in IEEE & 4-year degrees.

Electrical Engineering Outreach Seminar — Apr. 13, 2021
- Organized & hosted a seminar with NSBE where IEEE gave an overview of an education and career in electrical engineering to Black high schoolers; Created an interactive demo.

Halloween Toy Workshop — Oct. 20, 2020
- Proposed, coordinated, & executed a workshop where Purdue IEEE built toys proposed by elementary school children from rural Boswell, IN to demonstrate the engineering process.

HARDWARE SKILLS

- Mixed-Signal Design & Layout (28 nm, 180 nm, 350 nm)
- EEPROM (FG) Design, Programming, & Characterization
- Automated Test &Measurement (Analog Post-Si Verification)
- PCB Schematic Design and Layout (IPC Class II)
- FPGA | Signal Proc. | Semiconductors | Circuit Macromodels

SOFTWARE SKILLS

- Cadence Virtuoso
- Cadence Spectre & Spice
- Matlab, Python, &C (Console&Ebed)
- Altium Designer & KiCad
- Linux Environments

MISCELLANEOUS

Affiliations: NH & Boston IEEE Chapters (former), Purdue IEEE (former), Purdue iGEM (former)
Languages: English (Native), Bengali (Native), Hindi (Advanced), Japanese (Basic), German (Basic)
Music: Hindustani Classical Vocal Music, Violin, Bamboo Flute
Sports: Hiking (fully hiked the NH48), Taekwondo (2nd degree Black Belt), Running, Tennis, Soccer